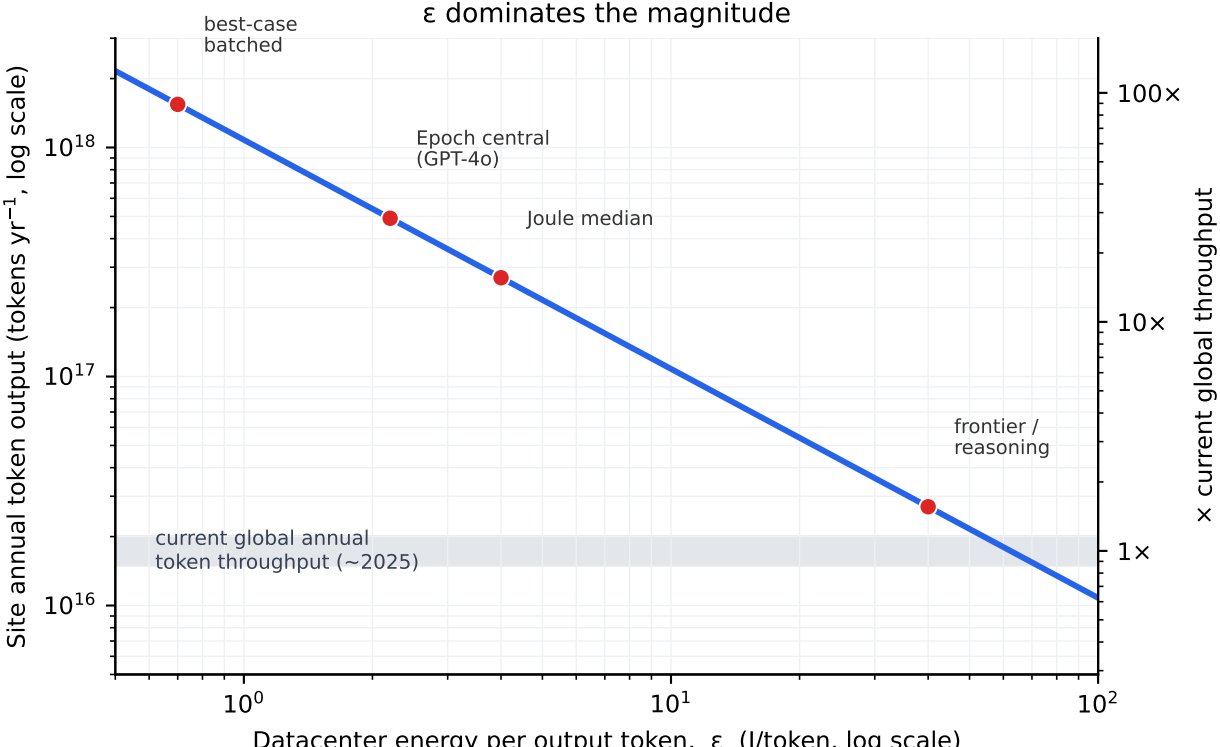
Fig. 3 | One ~300 TWh/yr site spans ~0.5× to ~30× global token demand; ε dominates the magnitude
best-case batched
Epoch central (GPT-4o)
Joule median
frontier / reasoning
current global annual token throughput (~2025)
Site annual token output (tokens yr^{-1}, log scale)
10^{18}
10^{17}
10^{16}
100×
10×
1×
× current global throughput
10^{0}
10^{1}
10^{2}
Datacenter energy per output token, ε (J/token, log scale)